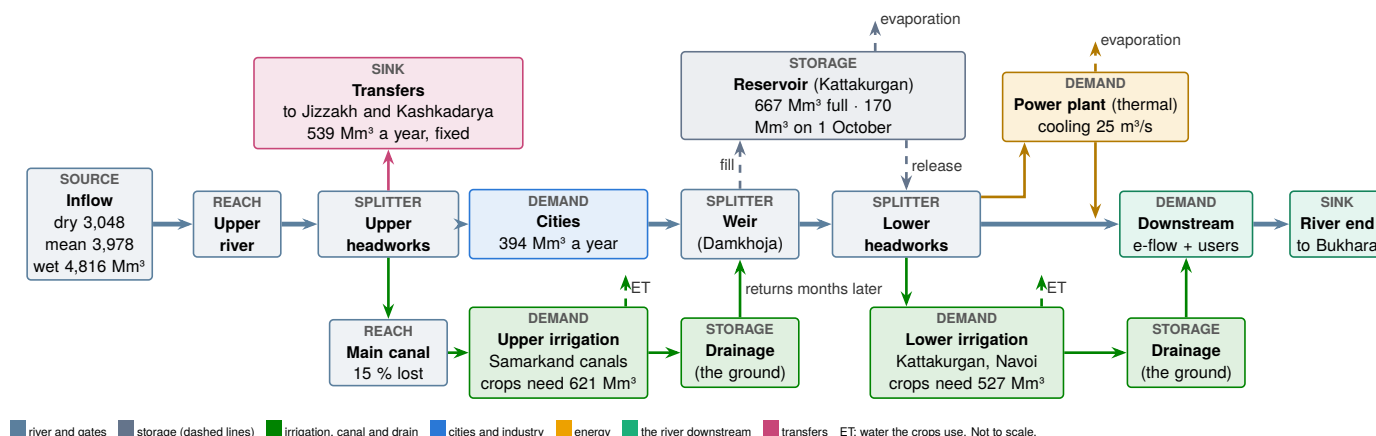


The Zarafshan in sixteen blocks: a first model of the oasis

Sheet for participants - TaqSim workshop, Tashkent, 12–13 October 2026

What you do. In 15 minutes you run a simplified Zarafshan river for one hydrological year, October to September, in monthly steps, in a dry, a mean or a wet year, or in the mean year as the climate models project it for 2041–2070 and 2071–2099. It is a game, not a test; all numbers are approximate. The game is one file and needs no internet.

The river as blocks



In each month every block, from upstream to downstream, takes what arrives, applies its rule and passes the rest on.

Kind	On a real river	What it does in each month	In this game
Source	where water enters	passes on the next value of its inflow series	Inflow, the river at Ravatkhoja
Reach	a river or canal stretch	can delay the water and lose part of it	Upper river, Main canal
Splitter	a weir or headworks	divides the water by a rule that you write; stores and loses nothing	Upper headworks, Weir, Lower headworks
Storage	a reservoir; the ground	stores water, can lose some, releases by a rule that you write	Reservoir, two Drainage blocks
Demand	a water user	takes what it needs, consumes part, returns the rest	Irrigation (two), Cities, Power plant, Downstream
Sink	where water leaves	records what arrives	Transfers, River end

Irrigation is two blocks, as on the river: the Samarkand canals above the reservoir, and the Kattakurgan command with Navoi below it; the upper fields come first. The reservoir lies beside the river between them, as Kattakurgan does: its water goes to the lower headworks. Drain water returns months later and fills the reservoir in winter.

Three levers: each is a number in the rule of a block

Lever	Range (start)	What it does
River reserve	0–40 % of inflow (15 %)	Each canal must leave at least this share of the month's natural inflow in the river (the power plant and the winter fill come first). More reserve: more water in the river, less for the crops of both blocks
Reservoir fill share	0–80 % (40 %)	From November to March the weir lets the power plant's intake pass and stores this share of the rest. The reservoir gives it back in summer
Reservoir release share	0–100 % (100 %)	Share of what is missing at the lower headworks (power plant, lower canal, reserve) that the reservoir gives, from March to November. A lower share keeps water for next year

Four results

Each has a tile at the top of the page. Higher is better on all four.

- **Upper irrigation** and **Lower irrigation** (%): the share of their crop requirement that the crops of each block receive.
- The **power plant** (thermal) has no tile: it gets at least 15 m³/s of cooling water in every month. Below 25 m³/s it makes less electricity; the small tile Energy not generated shows how much.
- **Downstream needs** (months of 12): months in which the river leaving the model carries the environmental flow (e-flow, 20 % of the natural inflow) plus the users' water.
- **End storage** (Mm³): water in the reservoir on 30 September. Below 67 Mm³ (dead storage) it cannot be released; it can still evaporate.

On a tile, ↑ means better and ↓ worse than the starting plan in the same year. The small bar shows your plan in all five worlds.

How to work

- **Task line** (top): six tasks and one extra, in order, each with a **Hint**. A task is ticked a moment after you let go of the slider. A note then says what the task showed; **Got it** or **Esc** closes it.
- **Year buttons** (top): dry, mean or wet year, or the mean year under climate change (2041–2070, 2071–2099). **Play the year** shows the months one by one.
- **Click a block** to see its kind, its rule, its water balance and a sketch of its TaqSim code.
- **Save plan** stores the lever positions under a name. **Starting plan** resets the levers.
- A check mark means supplied. A cross means not fully supplied; it does not mean that a task failed.

The six tasks. (1) Click one block of each kind. (2) Play the year. In the dry year: (3) meet the downstream needs in all 12 months; (4) give the lower fields at least 40 %; (5) keep 170 Mm³ in the reservoir. (6) In the Trade-off explorer, keep only the dry-year plans that do (3) and (5), and see how high they reach on Lower irrigation. Then find a mean-year plan that does all three. (7, extra) Switch to the 2071–2099 world and play the year: when does most water arrive now, and who pays?

How to read the Trade-off explorer

All 14,637 lever combinations were run in TaqSim in every world. The plot shows the plans that no other plan beats: 76 in the dry year, 63 in the mean year, 88 in the wet year, 64 and 60 in the two future worlds.

- **Four vertical axes**, one for each result. Higher is better on every axis. **Each grey line is one plan.**
- **Thick black line:** your plan now (faded when it fails a filter).
- **Numbered circles:** your saved plans.
- **Beaten.** Plan P beats plan Q when P is at least as good on all four results and better on at least one.
- **No line is the best.** Two lines that cross show a trade-off. Choosing among them is a decision, not a calculation.
- **Filter:** drag along an axis to keep a range; the lines that pass stay dark. On Downstream needs a filter keeps whole months. **Clear filters** removes it. **Load:** click a grey line to load that plan into the levers (not in step 1 of Task 6).

Words. Mm³: million cubic metres; 100 Mm³ in a month is about 38 m³/s. On lines: Mm³ that flowed; in a Storage: Mm³ held.